

e-salaries-for-different-job-roles

July 2, 2024

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
sns.set(color_codes=True)
```

```
[2]: #load the Dataset
df=pd.read_csv(r"C:\Users\dutta\Downloads\ds_salaries.csv")
```

```
[3]: df
```

```
[3]:      Unnamed: 0  work_year  experience_level  employment_type  \
0              0        2020                MI                FT
1              1        2020                SE                FT
2              2        2020                SE                FT
3              3        2020                MI                FT
4              4        2020                SE                FT
..          ...        ...                ...                ...
602          602        2022                SE                FT
603          603        2022                SE                FT
604          604        2022                SE                FT
605          605        2022                SE                FT
606          606        2022                MI                FT

      job_title  salary  salary_currency  salary_in_usd  \
0      Data Scientist    70000            EUR          79833
1  Machine Learning Scientist  260000            USD          260000
2      Big Data Engineer    85000            GBP          109024
3  Product Data Analyst    20000            USD           20000
4  Machine Learning Engineer  150000            USD          150000
..          ...        ...                ...                ...
602      Data Engineer  154000            USD          154000
603      Data Engineer  126000            USD          126000
604      Data Analyst   129000            USD          129000
605      Data Analyst   150000            USD          150000
606      AI Scientist   200000            USD          200000
```

	employee_residence	remote_ratio	company_location	company_size
0	DE	0	DE	L
1	JP	0	JP	S
2	GB	50	GB	M
3	HN	0	HN	S
4	US	50	US	L
..	...	...	...	...
602	US	100	US	M
603	US	100	US	M
604	US	0	US	M
605	US	100	US	M
606	IN	100	US	L

[607 rows x 12 columns]

```
[4]: # the first 5 rows of the Dataset
df.head()
```

```
[4]: Unnamed: 0  work_year  experience_level  employment_type  \
0          0      2020             MI          FT
1          1      2020             SE          FT
2          2      2020             SE          FT
3          3      2020             MI          FT
4          4      2020             SE          FT

      job_title  salary  salary_currency  salary_in_usd  \
0  Data Scientist   70000             EUR        79833
1  Machine Learning Scientist  260000             USD       260000
2  Big Data Engineer   85000             GBP       109024
3  Product Data Analyst  20000             USD        20000
4  Machine Learning Engineer  150000             USD       150000

employee_residence  remote_ratio  company_location  company_size
0                DE              0                DE              L
1                JP              0                JP              S
2                GB             50                GB              M
3                HN              0                HN              S
4                US             50                US              L
```

```
[5]: # the first 5 rows of the Dataset
df.tail()
```

```
[5]: Unnamed: 0  work_year  experience_level  employment_type  job_title  \
602        602      2022             SE          FT  Data Engineer
603        603      2022             SE          FT  Data Engineer
604        604      2022             SE          FT  Data Analyst
```

605	605	2022	SE	FT	Data Analyst
606	606	2022	MI	FT	AI Scientist

	salary	salary_currency	salary_in_usd	employee_residence	remote_ratio	\
602	154000	USD	154000	US	100	
603	126000	USD	126000	US	100	
604	129000	USD	129000	US	0	
605	150000	USD	150000	US	100	
606	200000	USD	200000	IN	100	

	company_location	company_size
602	US	M
603	US	M
604	US	M
605	US	M
606	US	L

```
[6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 607 entries, 0 to 606
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Unnamed: 0            607 non-null   int64
1   work_year             607 non-null   int64
2   experience_level      607 non-null   object
3   employment_type       607 non-null   object
4   job_title            607 non-null   object
5   salary               607 non-null   int64
6   salary_currency      607 non-null   object
7   salary_in_usd        607 non-null   int64
8   employee_residence   607 non-null   object
9   remote_ratio         607 non-null   int64
10  company_location     607 non-null   object
11  company_size         607 non-null   object
dtypes: int64(5), object(7)
memory usage: 57.0+ KB
```

```
[7]: # Descriptive Statistics of the Dataset
df.describe()
```

```
[7]:
```

	Unnamed: 0	work_year	salary	salary_in_usd	remote_ratio
count	607.000000	607.000000	6.070000e+02	607.000000	607.000000
mean	303.000000	2021.405272	3.240001e+05	112297.869852	70.92257
std	175.370085	0.692133	1.544357e+06	70957.259411	40.70913
min	0.000000	2020.000000	4.000000e+03	2859.000000	0.000000

25%	151.500000	2021.000000	7.000000e+04	62726.000000	50.000000
50%	303.000000	2022.000000	1.150000e+05	101570.000000	100.000000
75%	454.500000	2022.000000	1.650000e+05	150000.000000	100.000000
max	606.000000	2022.000000	3.040000e+07	600000.000000	100.000000

```
[8]: # check for duplicates
duplicates = df.duplicated()
```

```
[9]: # checking for null values
null_counts= df.isnull().sum()
if null_counts.sum()>0:
    print(null_counts[null_counts>0])
```

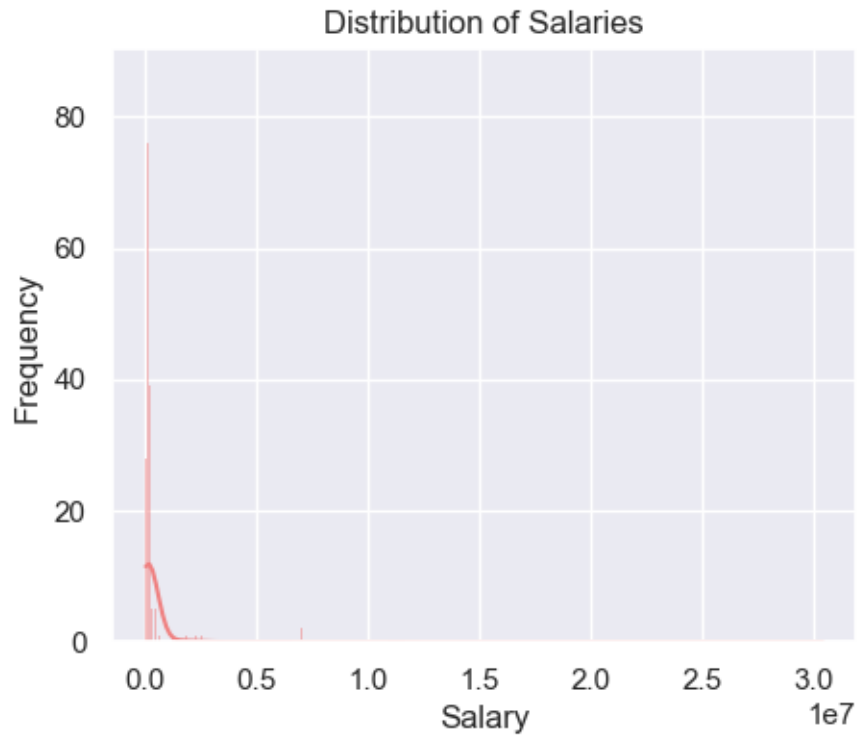
```
[10]: print(df.isnull().sum())
```

```
Unnamed: 0      0
work_year      0
experience_level 0
employment_type 0
job_title      0
salary         0
salary_currency 0
salary_in_usd   0
employee_residence 0
remote_ratio    0
company_location 0
company_size    0
dtype: int64
```

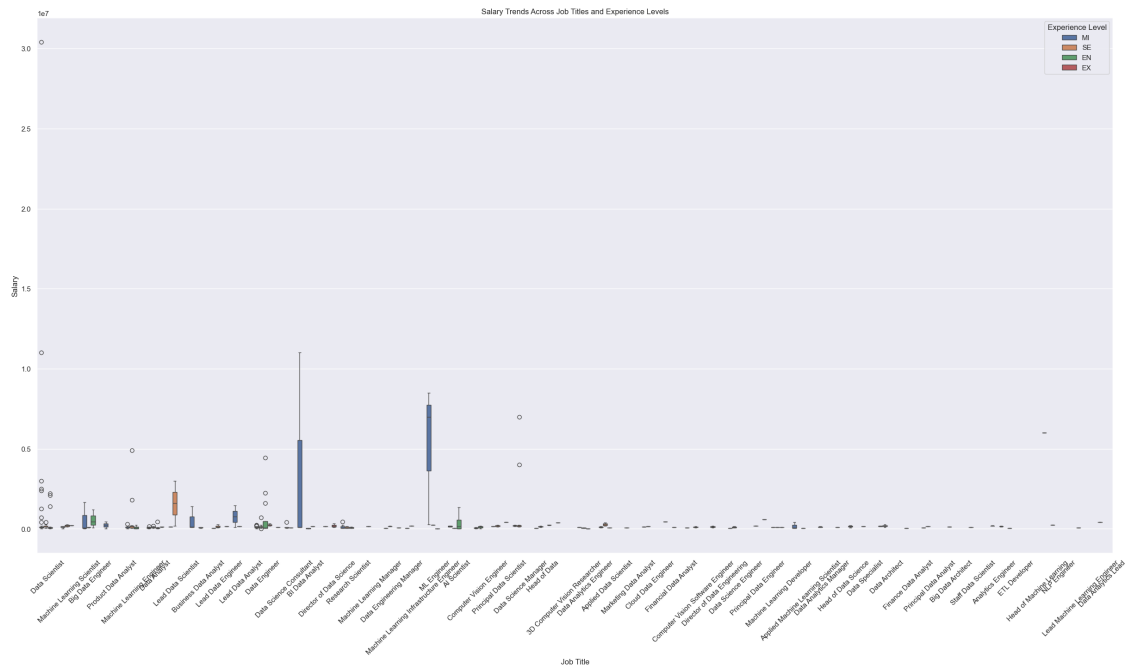
```
[11]: df.columns
```

```
[11]: Index(['Unnamed: 0', 'work_year', 'experience_level', 'employment_type',
        'job_title', 'salary', 'salary_currency', 'salary_in_usd',
        'employee_residence', 'remote_ratio', 'company_location',
        'company_size'],
        dtype='object')
```

```
[14]: # Histogram
plt.figure(figsize=(5,4))
sns.histplot(df['salary'], kde=True, color='lightcoral')
plt.title('Distribution of Salaries')
plt.xlabel('Salary')
plt.ylabel('Frequency')
plt.show()
```

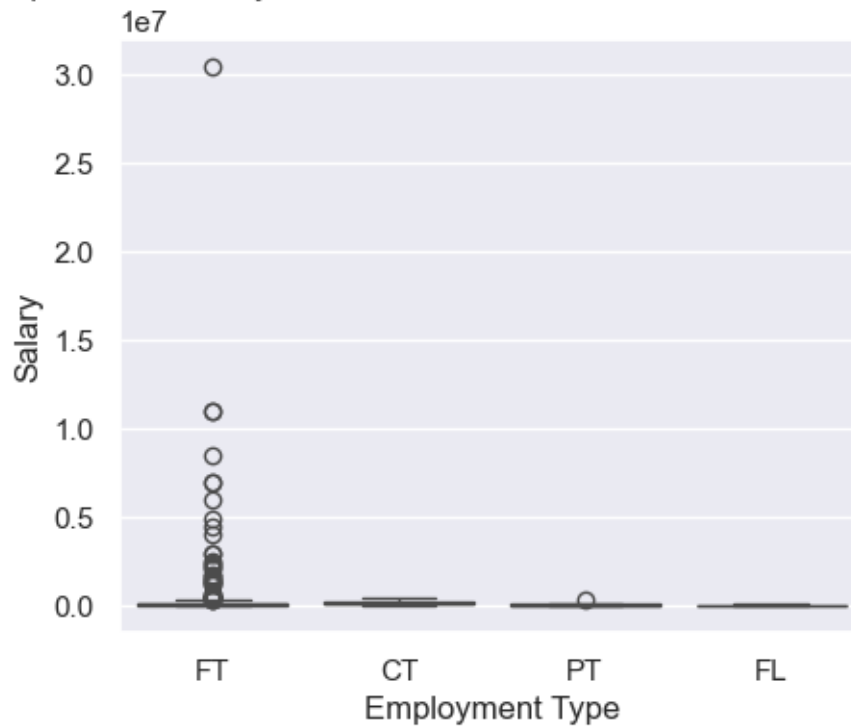


```
[15]: # Box plot
plt.figure(figsize=(30, 15))
sns.boxplot(x='job_title', y='salary', hue='experience_level', data=df)
plt.title('Salary Trends Across Job Titles and Experience Levels')
plt.xlabel('Job Title')
plt.ylabel('Salary')
plt.xticks(rotation=45)
plt.legend(title='Experience Level')
plt.show()
```

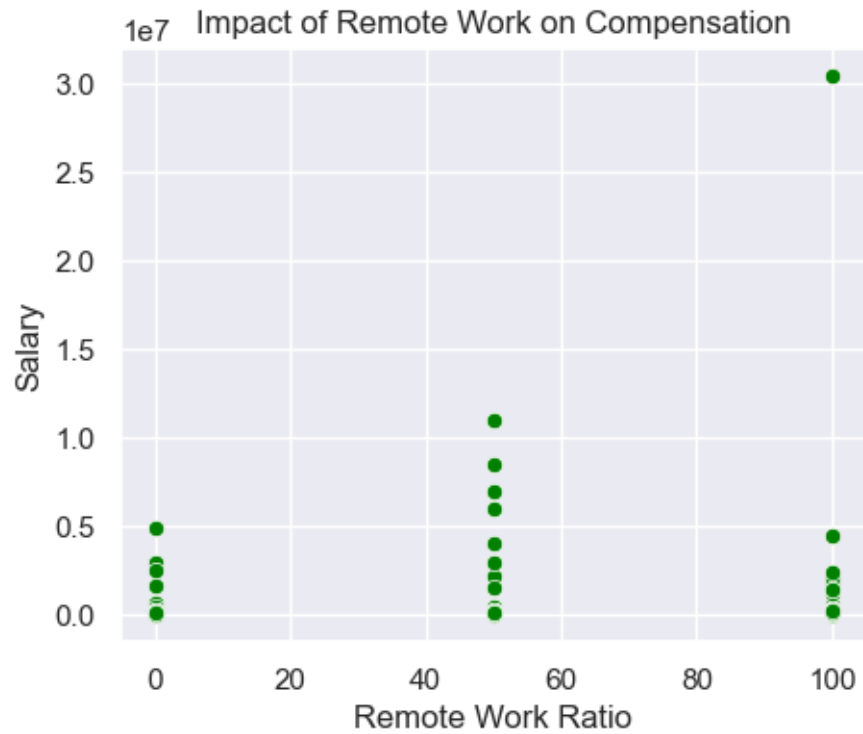


```
[17]: # Box plot
plt.figure(figsize=(5, 4))
sns.boxplot(x='employment_type', y='salary', data=df)
plt.title('Comparison of Salary Levels between Full-time and Part-time_
↳ Employment')
plt.xlabel('Employment Type')
plt.ylabel('Salary')
plt.show()
```

Comparison of Salary Levels between Full-time and Part-time Employment



```
[18]: # Scatter plot
plt.figure(figsize=(5, 4))
sns.scatterplot(x='remote_ratio', y='salary', data=df, color='green')
plt.title('Impact of Remote Work on Compensation')
plt.xlabel('Remote Work Ratio')
plt.ylabel('Salary')
plt.show()
```

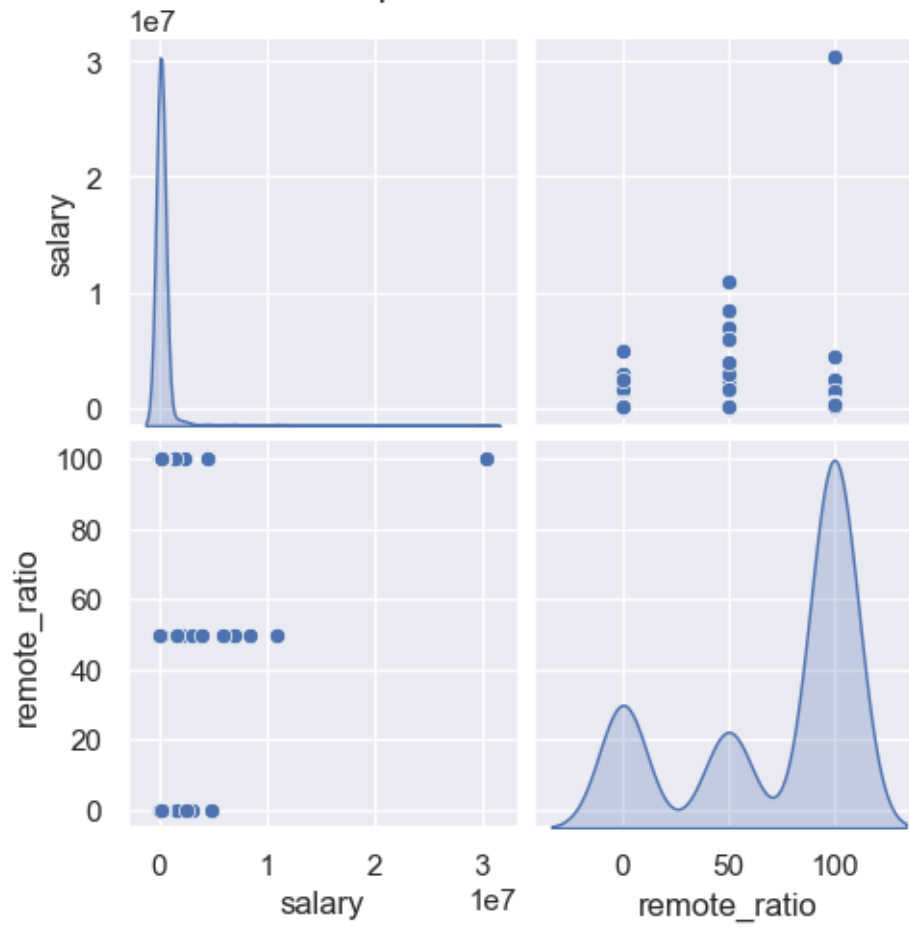


```
[19]: # Bar plot
plt.figure(figsize=(5,4))
sns.barplot(x='company_size',y='salary',data=df)
plt.title('Correlation between Company Size and Employee Salaries')
plt.xlabel('Company Size')
plt.ylabel('Average Salary')
plt.show()
```



```
[20]: # pair plot
sns.pairplot(df[['salary', 'remote_ratio', 'company_size', 'experience_level']],
            diag_kind='kde')
plt.suptitle('Pairwise Relationship between Numerical Variables', y=1.02)
plt.show()
```

Pairwise Relationship between Numerical Variables



[]: